The due date for submitting this assignment has passed.
Due on 2026-04-08, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-08, 08:45 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ):

1) An inner product on R^3 is defined as: ***1 point***

$$\langle .,. \rangle : \mathrm{R}^3 \times \mathrm{R}^3 \rightarrow \mathrm{R}$$

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + x_2y_2 + x_3y_3.$$

Match the sets of vectors in column A with their properties of orthogonality or orthonormality in column B with respect to the above inner product.

	Set of vectors (Column A)		Properties (Column B)
a)	$\{(2,3,4),(-1,2,-1)\}$	i)	Forms a basis but not orthogonal
b)	$\{\frac{1}{\sqrt{2}}(1,0,-1), \frac{1}{\sqrt{2}}(-1,0,-1)\}$	ii)	Forms an orthogonal basis
c)	$\{(2,3,4),(-1,2,-1),(0,4,-3)\}$	iii)	Orthogonal but not orthonormal, and does not form a basis of $\mathbb{R}^3$
d)	$\{(2,3,4),(-1,2,-1),(11,2,-7)\}$	iv)	Orthonormal, but does not form a basis of $\mathbb{R}^3$

Table : M2W8G1

Choose the set of correct options.